Response: HVarPart relative importance

Response

We thank the reviewer for identifying that the previous post-processing did not preserve negative adjusted hierarchical-partitioning contributions. We have replaced that pathway with a single extractor that retains HVarPart's `Unique`, `Average.share`, `Individual`, `I.perc(%)`, and total adjusted

R^2 values without truncation.

Our primary estimand is now the **signed allocation of adjusted explained variation**. Within reporting group g, predictor p is pooled as

$$A_{p,g} = \frac{\sum_m I_{p,m}}{\sum_m R^2_{\mathrm{adj},m}},$$

where $I_{p,m}$ is HVarPart's unmodified individual contribution. Models are excluded only when required predictors are absent, the total adjusted R^2 is non-finite or non-positive, or individual contributions are non-finite. Every exclusion is recorded in `Outputs/Tables/HVarPart/hvarpart_model_audit.csv`.

A negative allocation is an adjusted partitioning result, often associated with shared information, suppression, collinearity, weak information, or sampling variation. It is **not** evidence of a negative ecological effect.

For spatial SPD, the signed pooled allocation is 76.4% for climate and 23.6% for human predictors. The exact-zero sensitivity gives 75.6% climate, while excluding complete models with any negative contribution gives 72.4% climate. Across all five analyses, climate remains larger than human under all three profiles: yes.

Table 1: Model-level HVarPart eligibility audit. Counts are not duplicated across predictor rows.

analysis	n_models	n_eligible	n_excluded	n_negative_unique	n_negative_individual
h2_spd	31	31	0	1	0
spatial_events	1262	257	1005	75	34
spatial_spd	1262	1118	144	481	265
temporal_events	85	76	9	13	10
temporal_spd	85	84	1	17	9

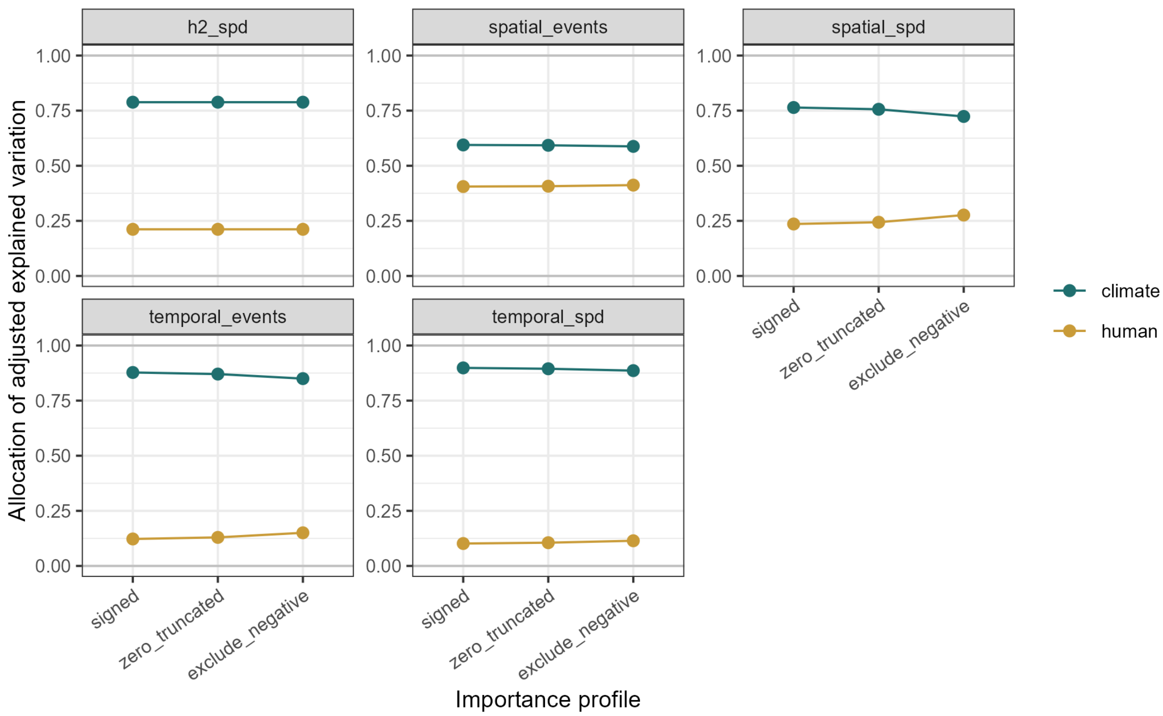

Figure 1: Signed primary allocation and two sensitivity profiles. Values are pooled from unmodified fitted HVarPart results.

Figure changes

Figure 2 has first been recreated with its region-map and predictor-panel composition retained. It now labels the estimand “Signed allocation of adjusted explained variation”. The central view uses shared global 1st–99th percentile limits rounded outward; boundary symbols and counts identify values outside those limits. No value is removed from the exported data, and a full-range supplementary rendering is provided.

Temporal H1 and H2 importance displays are unstacked and include explicit zero and one reference lines. The established cores 14944 and 15394 show HVarPart's original percentages and label total adjusted R^2 explicitly.

Reconciliation

Table 2: Programmatic reconciliation of formulas, exclusions, figure values, and artifacts.

check	passed
unique_model_predictor_keys	TRUE
signed_direct_formula	TRUE
paired_signed_allocations_sum_to_one	TRUE
audit_exclusions_reconcile	TRUE
figure2_values_match_components	TRUE
climate_larger_under_all_profiles	TRUE
required_artifacts_exist	TRUE

All numerical statements in this response are computed from `hvarpart_profile_comparison.csv`; audit statements use `hvarpart_model_audit.csv`; and reconciliation uses `hvarpart_validation_provenance.c`